

REVENUE CYCLE INTELLIGENCE

# ClaimFlow — Revenue Cycle & Claims Recovery Analytics

Helping healthcare providers recover denied claims and optimize cash flow through intelligent, automated analytics.

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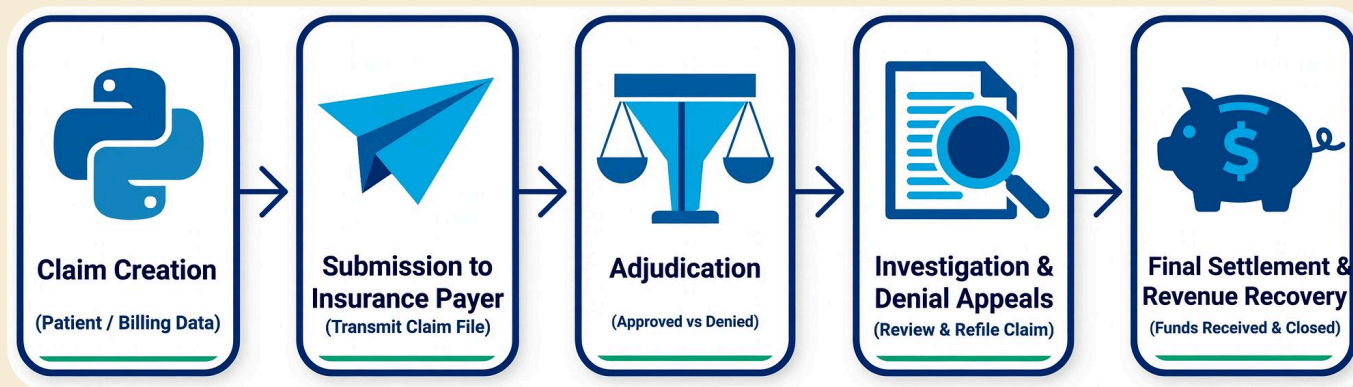


# The Revenue Recovery Challenge

## Why This Matters

Healthcare providers lose millions annually to claim denials and delayed settlements. Billing teams lack real-time visibility into which payers deny claims and why revenue stalls.

- Enterprise-mimicked dataset reflecting true settlement logic
- Root cause: no payer-level denial visibility
- Strategic goal: automated analytics for revenue recovery



# Four Business Questions, Answered

1

## Payer Risk

Which insurance payers deny the most claims, and what are the primary denial reasons?

2

## Aging Cash

How much revenue is stuck in pending claims, and what is its age distribution?

3

## Facility Bottlenecks

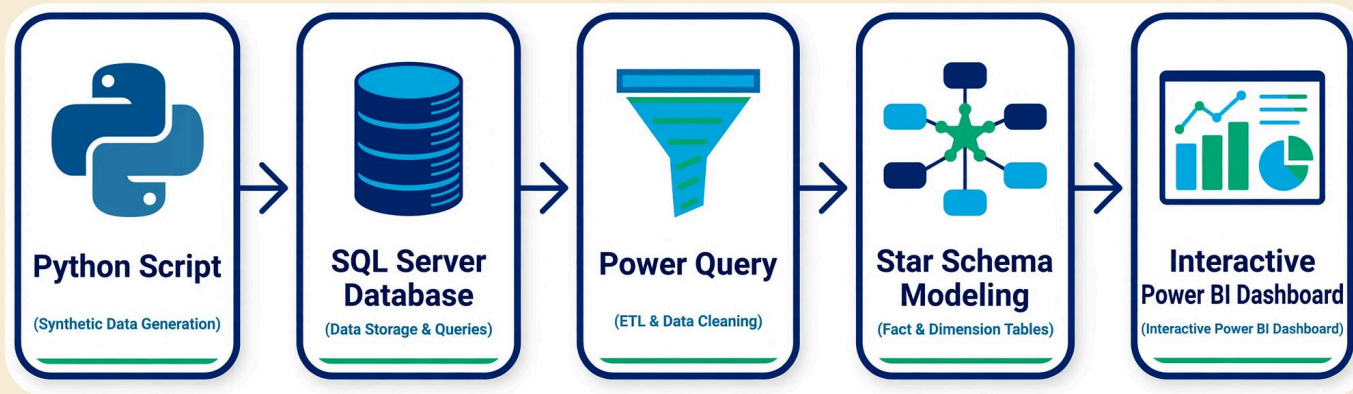
Which hospital facilities have the highest denial rates and operational delays?

4

## Financial ROI

Is it financially worth appealing denied claims, and how much revenue can be recovered?

# Technical Workflow & Architecture



## End-to-End Pipeline

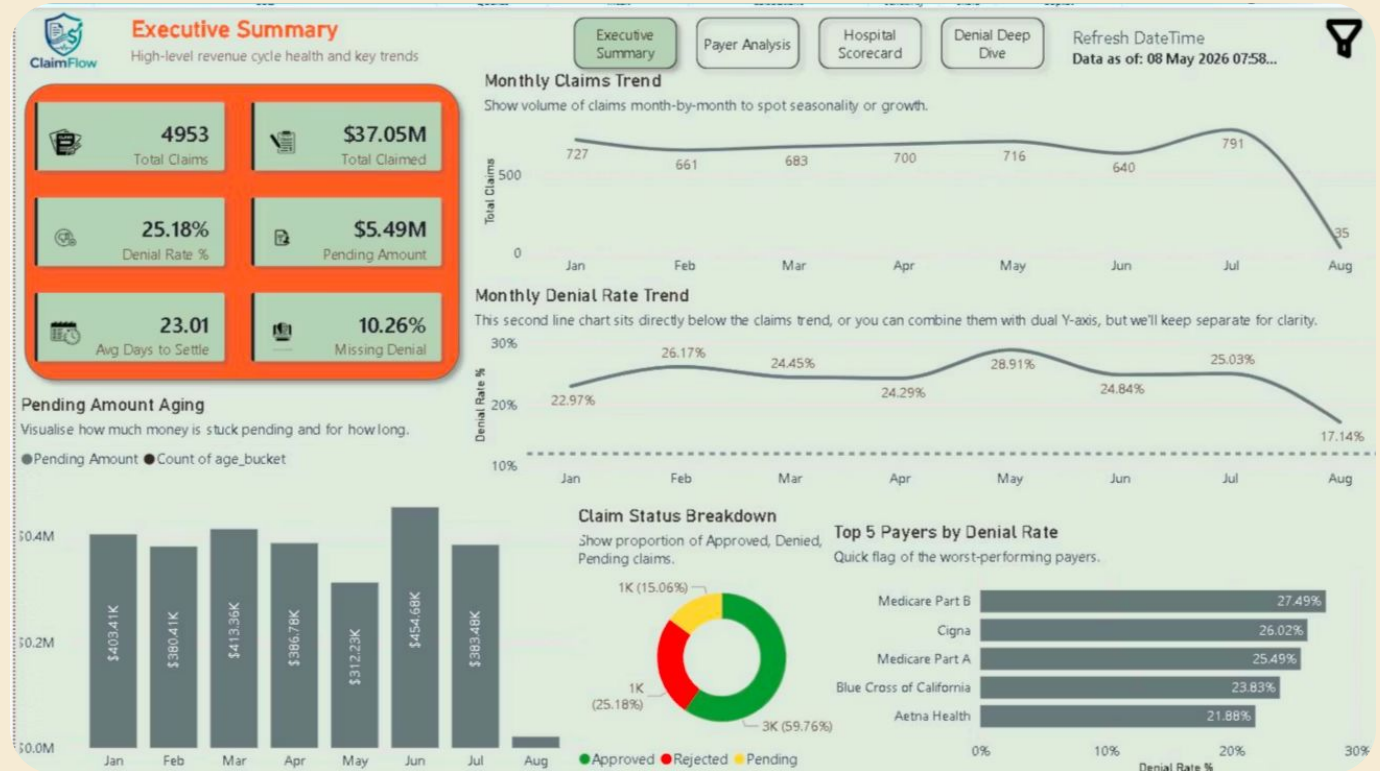
- **ETL & Data Engineering:** Robust cleaning, deduplication, and logic transformations via SQL Server and Power Query
- **Data Modeling:** Optimized Star Schema — 1 Fact Table (fact\_claims) connected to 3 Dimension Tables (dim\_hospitals, dim\_payers, dim\_date)
- **Tech Stack:** Python »→ SQL Server »→ Power Query »→ Power BI

# Executive Dashboard: 360° Financial Snapshot

## What Leadership Sees

A high-level overview of revenue cycle health — enabling instant, data-driven decisions without manual reporting.

- Total Claims Volume & Billed Amount
- Overall Denial Rate (%)
- Total Pending Revenue
- Average Settlement Days
- Monthly Claim Volume Trends

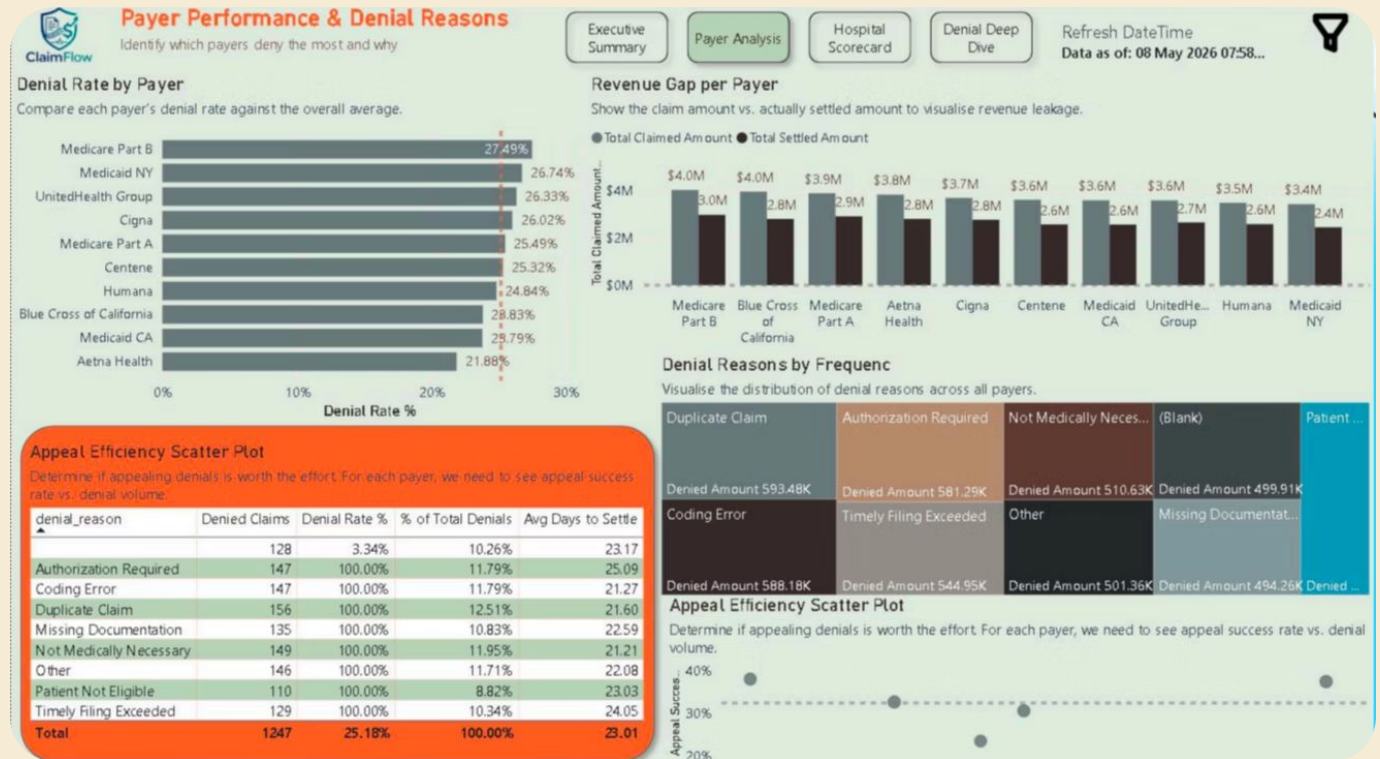


# Payer Analytics & Revenue Leakage

## Pinpointing High-Risk Payers

Identify insurance payers driving maximum financial leakage — exposing contractual non-compliance and strict documentation requirements before they compound.

- Denial Rate by Payer
- Billed vs. Settled Revenue Gap
- Denial Reason Treemap Distribution





# Facility & Operational Performance



## Locating Operational Bottlenecks

Evaluate hospital locations to isolate facility-level delays from external payer issues — turning operational blind spots into targeted improvement opportunities.

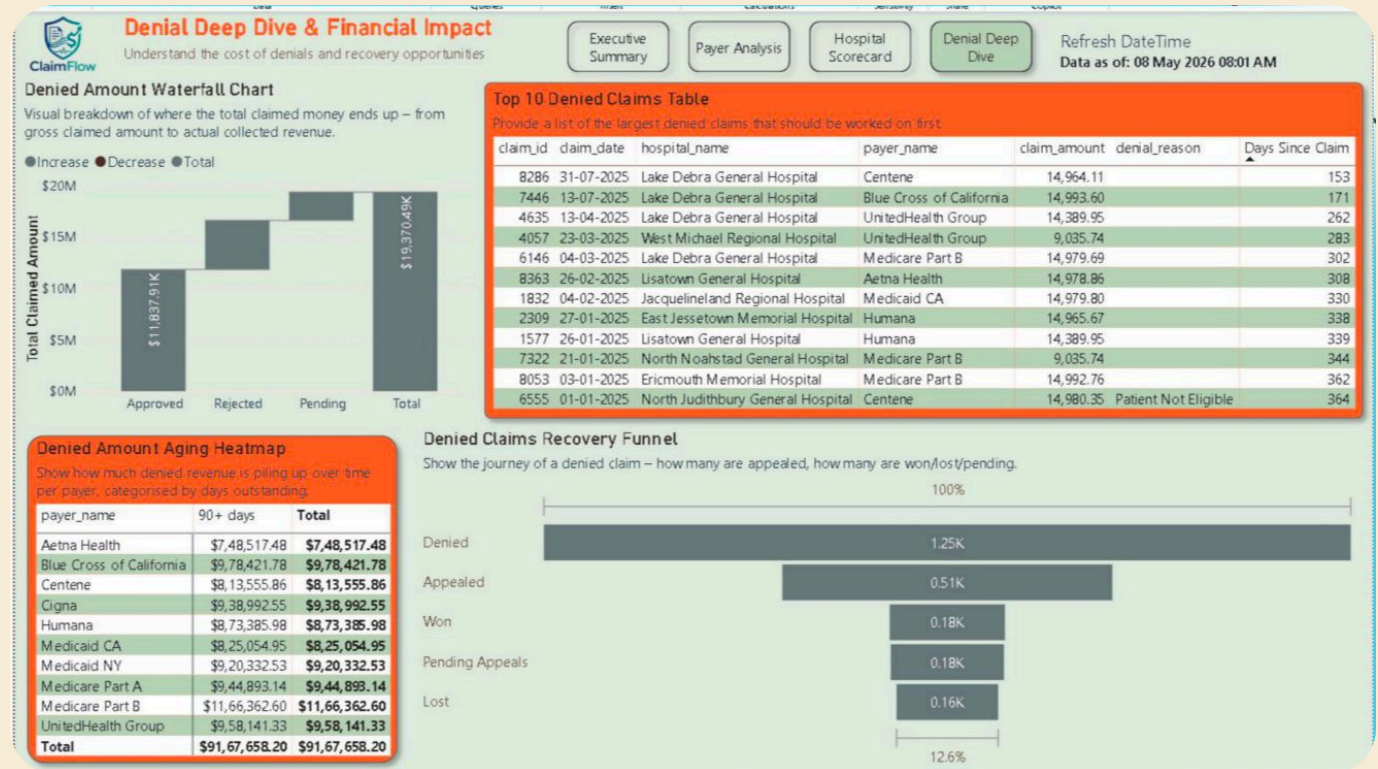
- Geographic Denial Map
- Hospital Comparison Scorecard
- Claim Volume vs. Denial Rate Scatter Plot

# Financial Impact & Recovery Deep-Dive

## Quantifying Recovery Potential

Prioritize high-value denied claims for immediate billing team intervention — converting aging receivables into recovered revenue.

- Claims Waterfall Recovery Chart
- 30-60-90+ Days Aging Heatmap
- Appeals Success Funnel





## Key Insights & Strategic Recommendations

## Payer Strategy

Commercial payers deny **15–18%** of claims, primarily due to "Missing Documentation." **Action:** Implement automated pre-submission data validation.

# Aging Management

Over ~\$450K is stuck in pending status, with **20%** exceeding 90 days.  
**Action:** Trigger automated alerts for claims pending >60 days.

## Appeals Optimization

Appeals recover **50%** of denied revenue, yet only **40%** are refiled.  
**Action:** Standardize appeal templates and mandate filing for claims >\$1,000.







# How I Can Drive Value for Your Organization



## Interactive Business Intelligence

Enterprise Power BI, Tableau & Looker Studio dashboards with responsive Desktop & Mobile views.



## Advanced Spreadsheet Solutions

Dynamic Excel/Google Sheets financial models, Google Apps Script automation, and AppSheet applications.



## Database Architecture & SQL

Relational database setup, Star Schema data modeling, complex SQL queries, and ETL pipeline design.



## Python Analytics

End-to-end data extraction, web scraping, data cleaning (Pandas/NumPy), and automated statistical reporting.



Looking for a dedicated data analyst to optimize your business reporting or solve revenue bottlenecks? Let's connect!

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